Retrocausal Signalling, Photonic Wormhole Geometry and Informatic Time Machines: A Speculative Framework

Working paper — hypotheses and proposed experiments

Abstract. This paper collects a set of speculative hypotheses concerning the nature of light, time and consciousness, and proposes a series of table-top experiments. The central conjecture is that entanglement-linked signals, measured classically over a large separation, may carry information to an earlier timestamp, implying that time is not absolute and that the universe does not all exist at the same instant. Supporting conjectures address photon structure, superoscillation, superposition-based propulsion, and the possibility of paradox-free time travel. A concrete 150 m fibre-optic test is described, together with a short section on known constraints and open problems. All claims are hypotheses; none has been experimentally verified.

1. Introduction

Time is treated here as non-linear and non-absolute. The core intuition is that if the speed of light is a universal feature of reality, a length of spacetime is fixed; communicating faster than light between two places would therefore deliver information to the past, with no time dilation between sender and receiver. If this can be realised, it would change interplanetary communication and open a route to what is termed an *informatic time machine*.

Truth is regarded as perceivable only through recurring patterns of nature, and the investigation of those patterns is taken to be the definition of intelligence.

2. Photons, Light Speed and Geometry

2.1 Faster-than-light transmission

It is proposed that photons possess a property permitting transmission of a force many times faster than *c* (a figure of 10,000× is suggested). A transmitter and receiver built on this effect would transform interplanetary communication.

2.2 Non-constant light speed and superoscillation

The speed of light is hypothesised not to be constant everywhere for a single observer. Superoscillation is proposed to open a wormhole-like corridor from the multiverse that carries energy, sometimes lying within our universe and sometimes in another. This corridor could be stabilised for the transfer of light. Teleportation would require entanglement together with the encoding of information.

2.3 Sub-photon structure and the double slit

Particles travelling at light speed through a double slit may separate from other particles and behave erratically until recombined elsewhere, suggesting photons are composed of smaller constituents. The wave packet may push empty space around itself at particular energy levels, negotiating transit for other wave packets. Any sensor moving at light speed would register distortions of existence resembling quantum background fluctuations, in which massive objects can disappear. Shortcomings in understanding the double slit are thought to block the development of particle-beam technology, including lasers that project mass at light speed.

2.4 Geometry without matter

Geometry can exist without matter. If photons are considered matter, they carry geometries even where they are not positioned, implying that spacetime itself has geometry. The *wormhole geometry* must exceed *c* (wg > c) to create a wormhole epoch without reverberation.

3. Time, Entanglement and Negative Energy

Time dilation of entangled photons between two points on alternate time-like curves is proposed to yield futuristic information, with possible application to multiverses and eventually AGI. Because time and velocity are mutual, entanglement is hypothesised to communicate to the past; the *wormhole epoch* is the temporal retrograde of entanglement.

The *EPR bridge* is proposed to convert energy into negative energy at other locations. Since energy cannot be destroyed, the universe must dilate time, and since entanglement outpaces light, time must flow backwards in proportion to the negative energy. Information can be encoded into negative energy, and negative energy can be amplified for greater time reversal. A related conjecture states that if pure energy converts to entropy, pure energy can also produce momentum. Memory performance is described as measured by disentanglement of entropy, and the past is held not to exist.

Retroactive precognition is expressed as:

$$RP = SCF + D$$

(retroactive precognition = spatial cognition forensics + distance), from which it is conjectured that deployable precognitive machines can be remembered.

4. Proposed Devices

4.1 Informatic time machine

Several Bose–Einstein condensates (BECs) are produced by laser cooling. Half of the masses undergo induced time dilation through high-speed transport. The dynamic masses are measured by time-division multiplexing on a specific environmental trigger, producing multiple anti-correlations. The static mass holds the information encoded across temporal partitions with those anti-correlations. Cross-referencing with statistical noise indicates the result is probably correct.

4.2 Chronotron

A discharge tube using cooled lithium gas as a BEC, accelerated by induction coils, to study time dilation. **Paradox mitigation:** a signal is sent only if the experiment succeeds, so a paradox arises only upon success. A further validity step: manipulate the random seed until a confirmation is received from the cache, and transmit only after confirmation.

4.3 Baryonic asymmetric force multiplier

Consists of a baryonic superposition waveguide, an inertial scoop and measurement apparatus intended to produce surplus momentum. Superpositional atoms are catapulted onto a metamaterial to generate the effect.

4.4 Möbius tape drive

A Möbius tape drive carries a program that spins the drive at an infinitely variable rate with programmable synchronous decumulation, retrieving encrypted information at program end after a set time. Programming involves producing a random seed, creating a key after the final result, then resetting the device.

4.5 Photoelectric wave-function-collapse experiment

Solar panels placed at different points within a wave function act as measurement devices through the photoelectric effect. In a double-slit setup, if a panel registers the photon, the beam is expected to collapse to a bilinear state and the photons to teleport. A light valve or resistor on the panel would enable or disable measurement, permitting “spooky action at a distance” to be toggled. A fallback is a light valve and polariser arrangement.

4.6 Hardware model for probabilistic decision (RP)

Figure 1 shows a hardware model of a randomized-polynomial (RP) decision process: two drivers (a witness channel and deterministic logic) feed an impulse into a witness-collector array and integrator, which clocks a 74HC74 flip-flop through a timer; the output gate selects a result with probability ≥ 1/2, mirroring the one-sided error bound of the RP complexity class. In this framework it illustrates how amplified impulses and clocked decision gates could serve as the measurement and verification stage of the time-machine apparatus.

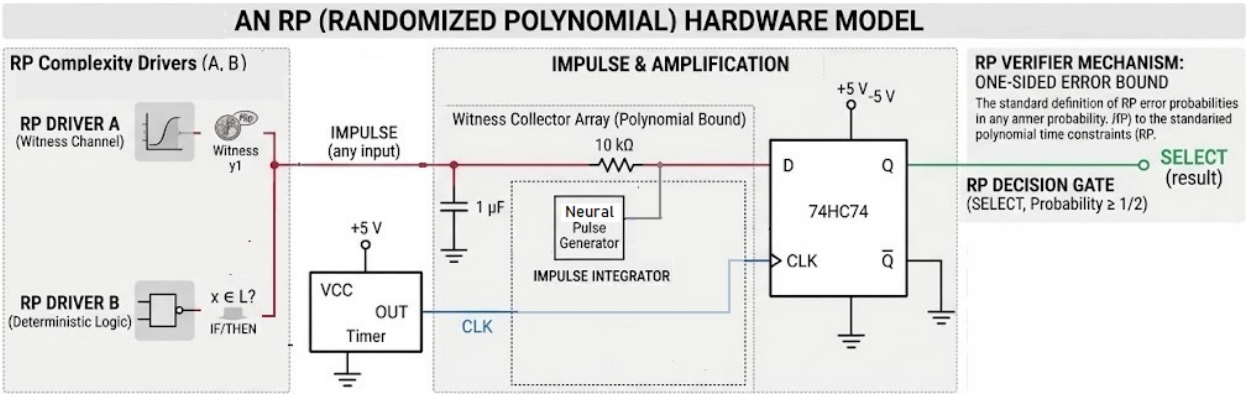


Figure 1. An RP (randomized polynomial) hardware model: complexity drivers, impulse and amplification stage, verifier mechanism and decision gate.

5. Proposed Experiment: 150 m Fibre-Optic Test

The time machine is hypothesised to transmit information into the past only with an extremely long fibre-optic cable whose end components are as far apart as possible. Light covers 100 m in approximately 0.333564095198 µs, which the author interprets as the temporal shift per 100 m. Using a sampling front-end with a 333 ns sample interval (a MAX11131-class ADC board, or an STM32F103C8T6 board as a low-cost alternative), a separation of about 150 m is required to collect sufficient data. The intended wormhole connects roughly 0.5 µs into the past across 150 m.

Components:

- 1×N ultra-fast optical switch (nanosecond delay)
- Fibre-optic inline polariser
- Multiple systems, to achieve causal drift

Procedure and checklist:

- Perform a test entanglement at each stage.
- Set up time-division multiplexing.
- Calibrate geometry for the unitary operator.
- Maximise the wormhole epoch.
- Set up an API for the results.

Continuity calibration is required for retrocausal time-division multiplexing of the exact experimental parameters, ensuring verifiable integrity of the quantum signal and information. Because the device cannot change the temporal destination once initiated, the system is expected to be paradox-free and to imply no parallel universe. Once the user alters the past, the past changes the distant present only, so a correctly positioned user should receive a signal from the future. Quantum entanglement cannot achieve this at a single location: a quantum signal measured classically over a large distance would carry a timestamp earlier than the start of the clock.

6. Consciousness

Time is not linear, and consciousness is hypothesised to survive short pulses of non-existence, so exploring the universe without a body may be possible. Long periods without a body would lead to oblivion; unless consciousness arrives at a body in time, nothing of the journey is remembered. Consciousness is further proposed to arise from dreams: a cloned person who dreams their memories would be conscious of themselves regardless of who was cloned. The author notes personal experience of Cotard's syndrome as motivation. Sending logic into the past is floated as a way to circumvent linear emotion, and it is suggested that time leads to apathy.

7. Computational Conjectures

A desktop machine performing 10^{10} operations per second might reach about 10^{14} operations per two seconds using subspace folding and deterministic computation mapping. If N = NP (the P = NP question), this could be used to mine cryptocurrency at roughly 50 TH/s regardless of algorithm. The author also suggests that a homogenic delta might improve thinking, and that a "petrichorial spacetime signature" emanates from paradoxes and teleportation.

8. Known Constraints and Open Problems

For the hypotheses above to be tested fairly, the following mainstream results must be addressed directly:

- **No-communication theorem.** Measurements on one half of an entangled pair do not change the other half's local statistics, so entanglement alone cannot carry a message, forward or backward in time. Any proposal must explain what additional mechanism circumvents this.
- **Light-travel time versus time dilation.** 0.3336 μs per 100 m is simply the time light takes to cross 100 m (100 / c). It is a propagation delay, not a temporal shift or dilation, and a 333 ns sampler will detect it in any ordinary fibre run. A control experiment is needed that would distinguish a genuine retrocausal timestamp from ordinary delay.
- **Relativity and causality.** Superluminal signalling leads to causal paradoxes in special relativity; the proposed paradox-mitigation rule (send only on success) selects outcomes after the fact and requires a precise physical justification.
- **Photon structure.** Photons are well described as massless excitations of the electromagnetic field; evidence for sub-photon constituents would need independent experimental support.
- **Computation claims.** The throughput and mining claims rest on N = NP, which is unproven and widely believed false; without it, no known method gives the stated speed-up.
- **Verification design.** Pre-registering the expected result, running blinded controls, and using independent clocks would allow the 150 m test to produce a meaningful negative or positive outcome.

9. Conclusion

The framework proposes that entanglement, wormhole-like geometry and time-division multiplexing could together permit information to arrive at an earlier timestamp. The first step is the 150 m fibre experiment described in Section 5, designed with the controls listed in Section 8. Until such data exist, the contents of this paper should be read as hypotheses.